KARABAKH UNIVERSITY

Faculty of Engineering • Department of Computer Engineering

Laboratory Report № 1: Conditional Statements in Python

Course:	Introduction to Programming (Python)	**Date:**	October 12, 2026
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1. Executive Summary & Learning Objectives

This laboratory assignment explores the practical application of conditional decision-making mechanisms in Python programming. By leveraging control structures such as `if`, `elif`, and `else`, along with arithmetic and logical operators, three distinct applications were developed to solve computational and logic-based problems.

- Implement decision-making logic using conditional branching statements.
- Apply arithmetic operations (exponentiation, modulo operator) to real-world scenarios.
- Implement set-based binary logic algorithms for interactive problem solving.
- Structure software projects following clean code principles and documentation guidelines.

2. Task 1: Body Mass Index (BMI) Calculator

Description: Computes Body Mass Index based on user-provided weight (kg) and height (m) parameters using the formula: **BMI = weight / (height × height)**.

BMI Range	Classification Category	Conditional Mapping
Below 18.5	Underweight	`bmi < 18.5`
18.5 – 24.9	Normal Weight	`18.5 <= bmi <= 24.9`
25.0 – 29.9	Overweight	`25.0 <= bmi <= 29.9`
30.0 and above	Obese	`else`

Python Implementation (Task1.py)

```
      weight = float(input("Enter weight (kg): ")) height = float(input("Enter height (m): ")) bmi = weight / (height * height) if bmi < 18.5: category = "Underweight" elif 18.5 <= bmi <= 24.9: category = "Normal Weight" elif 25.0 <= bmi <= 29.9: category = "Overweight" else: category = "Obese" print(bmi) print(category)
```

3. Task 2: Chinese Zodiac Calculator

Description: The Chinese Zodiac operates on a 12-year cyclic calendar where each year corresponds to a specific animal sign. The algorithm computes the remainder of the birth year divided by 12 (`year % 12`) and maps it to the respective animal using a dictionary data structure.

Python Implementation (Task2.py)

```
      birth_year = int(input("Enter your birth year: ")) c = birth_year % 12 d = { 0: "Monkey", 1: "Rooster", 2: "Dog", 3: "Pig", 4: "Rat", 5: "Ox", 6: "Tiger", 7: "Rabbit", 8: "Dragon", 9: "Snake", 10: "Horse", 11: "Sheep" } zs = d[c] print("Your chinese zodiac sign is: ", zs)
```

4. Task 3: Interactive Birthday Guessing Game

Description: A guessing algorithm that calculates the user's birth day (1–31) using five numerical sets based on binary positional representation (powers of 2: 1, 2, 4, 8, 16). For each set in which the birthday appears, the corresponding base weight is added to an accumulator variable.

Python Implementation (Task3.py)

```
      day = 0 print("Is your birthday in Set1?\n 1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31") if int(input("Enter 0 for No and 1 for Yes: ")) == 1: day += 1 print("\nIs your birthday in Set2?\n 2 3 6 7 10 11 14 15 18 19 22 23 26 27 30 31") if int(input("Enter 0 for No and 1 for Yes: ")) == 1: day += 2 print("\nIs your birthday in Set3?\n 4 5 6 7 12 13 14 15 20 21 22 23 28 29 30 31") if int(input("Enter 0 for No and 1 for Yes: ")) == 1: day += 4 print("\nIs your birthday in Set4?\n 8 9 10 11 12 13 14 15 24 25 26 27 28 29 30 31") if int(input("Enter 0 for No and 1 for Yes: ")) == 1: day += 8 print("\nIs your birthday in Set5?\n16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31") if int(input("Enter 0 for No and 1 for Yes: ")) == 1: day += 16 print("\nYour birthday is:", day)
```

5. Evaluation & Results

All three tasks were tested against sample inputs provided in the lab guidelines. The programs produced expected outputs without computational errors or logical faults. The modular structure and clear conditional branching ensured complete alignment with academic requirements.